Cyber Threat Intelligence (CTI) Strategy Report

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1. Objectives and Scope Definition

Define Objectives

SecureHealth's Cyber Threat Intelligence (CTI) strategy is designed to strengthen the organization's cybersecurity posture against increasing ransomware attacks targeting healthcare organizations. The primary goals include:

Enhance Threat Detection and Mitigation

- Improve the organization's ability to identify ransomware attacks before they spread.
- Detect Indicators of Compromise (IOCs) using proactive monitoring and intelligence feeds.
- Reduce attacker dwell time and lateral movement within the network.

Protect Patient Data and Maintain Compliance

- Secure Electronic Medical Records (EMRs) from unauthorized access and exfiltration.
- Ensure compliance with HIPAA and healthcare cybersecurity regulations.
- Monitor for stolen credentials and suspicious user activities.

Improve Incident Response and Recovery

- Strengthen containment and recovery capabilities during ransomware incidents.
- Reduce operational downtime affecting patient care services.
- Improve coordination between security and operational teams.

Enhance Threat Awareness

- Increase visibility into emerging ransomware groups, phishing campaigns, and healthcare-focused cyber threats.
- Support informed decision-making through strategic threat intelligence reporting.

Define Scope

The CTI strategy will apply to the following operational and technical areas within SecureHealth:

Departments Covered

- IT Security Team
- Incident Response Team
- Compliance Team
- Network Administration Team
- Executive Management

Systems and Infrastructure Covered

- Electronic Medical Record (EMR) systems
- Telehealth platforms
- Cloud-based healthcare applications
- Endpoints and servers
- Network infrastructure
- Vendor access systems
- Backup and recovery systems

Operational Boundaries

The CTI program will focus on:

- Internal organizational networks
- Remote employee access
- Third-party vendor connections
- Threats targeting healthcare operations
- Data privacy and regulatory compliance requirements

Policy Alignment

Several security policies align closely with SecureHealth's CTI goals.

Policy	Importance to CTI Strategy

Access Control Policy	Helps reduce unauthorized access and credential abuse.
Incident Response Policy	Guides detection, containment, eradication, and recovery procedures.
Data Protection and HIPAA Compliance Policy	Ensures patient data confidentiality and regulatory compliance.
Vendor Risk Management Policy	Addresses risks associated with third-party access to systems.
Backup and Recovery Policy	Supports resilience against ransomware attacks and operational disruptions.

These policies are priorities because they directly support ransomware prevention, patient data protection, and operational continuity. They also guide future CTI activities such as monitoring, incident response, and intelligence sharing.

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2. Threat Intelligence Needs Assessment

Risk Assessment

SecureHealth faces several high-priority cybersecurity threats due to the sensitive nature of healthcare operations.

Threat	Likelihood	Impact	Description
Ransomware Attacks	High	High	Malware encrypts systems and disrupts patient care operations.
Unauthorized Access	High	High	Stolen credentials may allow attackers access to EMR systems.
Phishing Attacks	High	Medium	Employees may unknowingly deliver malware or expose credentials.
Third-Party Vendor Risks	Medium	High	Vendors with system access may become attack entry points.

Data Breaches	Medium	High	Exposure of patient information may lead to compliance violations.

Key Vulnerabilities

- Weak password practices
- Lack of employee phishing awareness
- Unpatched systems and software
- Excessive user privileges
- Third-party access exposure

Stakeholder Collaboration

Feedback from organizational stakeholders helped identify intelligence requirements.

IT Security Team

Required Intelligence:

- Tactical intelligence
- Malware IOCs
- Threat actor Tactics, Techniques, and Procedures (TTPs)

Purpose:

- Improve EDR detection rules and monitoring.

Compliance Team

Required Intelligence:

- Strategic intelligence
- Regulatory threat reports
- Breach trends affecting healthcare

Purpose:

- Maintain HIPAA compliance and reduce legal risk.

Incident Response Team

Required Intelligence:

- Operational intelligence
- Real-time ransomware alerts
- Containment and recovery guidance

Purpose:

- Improve response speed and recovery effectiveness.

Executive Leadership

Required Intelligence:

- Strategic threat summaries
- Business risk assessments

Purpose:

- Support cybersecurity decision-making and budgeting.

Prioritization of Intelligence Needs

SecureHealth prioritizes intelligence needs based on threats with the highest likelihood and operational impact.

Highest Priority: Ransomware Intelligence

Ransomware is prioritized because:

- It directly disrupts patient care operations.
- Healthcare organizations are frequent ransomware targets.
- Downtime significantly impacts business continuity.

Second Priority: Credential Theft and Unauthorized Access

This is prioritized because:

- Attackers commonly exploit stolen credentials.
- EMR systems contain sensitive patient information.

Third Priority: Third-Party Risk Intelligence

Vendors are prioritized because:

- Third-party access expands the attack surface.
- Compromised vendors can bypass internal defenses.

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3. Gathering and Integrating Intelligence Sources

Intelligence Sources

Internal Sources

Source	Description
SIEM Logs	Centralized monitoring of security events and alerts.
Endpoint Logs	Detect suspicious endpoint activities.
Authentication Logs	Identify failed or unusual login attempts.
Incident Reports	Historical attack data for trend analysis.
Network Traffic Data	Monitor abnormal communications and malware activity.

External Sources

Source	Description
Recorded Future	Threat intelligence feeds and ransomware tracking.
Healthcare ISAC	Industry-specific healthcare threat intelligence sharing.
Open-Source Intelligence (OSINT)	Publicly available threat information and malware analysis.
Government Advisories	Alerts from CISA, FBI, and healthcare regulators.
Dark Web Monitoring	Detection of leaked credentials and threat actor discussions.

Integration Process

Threat intelligence is integrated into SecureHealth's security infrastructure through:

- Integration of Recorded Future feeds into SIEM platforms.
- Automated IOC ingestion into CrowdStrike EDR systems.
- Correlation of threat data with internal logs.
- Dashboard reporting for analysts and incident responders.

Tools Used

- Recorded Future Threat Intelligence Platform
- CrowdStrike EDR Platform
- SIEM monitoring systems
- Threat intelligence dashboards

Source Validation

SecureHealth validates intelligence sources through:

- Cross-referencing multiple intelligence feeds.
- Verifying source reputation and reliability.
- Reviewing historical accuracy of threat reports.
- Prioritizing intelligence from trusted government and industry sources.
- Removing outdated or irrelevant intelligence.

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4. Development of Threat Intelligence Processes

Data Collection Procedures

Threat data collection involves:

- Gathering Indicators of Compromise (IOCs)
- Monitoring endpoint telemetry
- Collecting firewall and SIEM logs
- Monitoring phishing emails and malicious domains
- Capturing malware hashes and suspicious IP addresses

Collection Techniques

- Automated threat feed ingestion
- Endpoint monitoring
- Network traffic analysis
- Threat hunting activities
- Dark web monitoring

Analysis and Insight Generation

Security analysts use several techniques to generate actionable intelligence.

Analysis Methods

- Trend analysis for ransomware activity
- MITRE ATT&CK framework mapping
- Behavioral analysis of malware
- Correlation analysis between multiple alerts
- Threat actor profiling

Key Focus Areas

- Emerging ransomware tactics
- Phishing attack trends
- Credential theft techniques
- Lateral movement patterns

Dissemination Protocols

Threat intelligence is shared according to the severity and audience.

Team	Dissemination Method
Security Operations Center (SOC)	Real-time alerts and dashboards
Incident Response Team	Immediate containment notifications
Compliance Team	Regulatory impact reports

Executive Management	Monthly strategic summaries

Reporting Frequency

- Critical alerts: Immediate
- Operational reports: Daily or weekly
- Strategic reports: Monthly or quarterly

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5. CTI Capability and Infrastructure Building

Tool and Platform Acquisition

Tool/Platform	Purpose
Recorded Future	Threat intelligence aggregation and analysis
CrowdStrike	Endpoint detection and response
Veeam	Backup and disaster recovery
SIEM Platform	Security monitoring and event correlation
MFA Solutions	Protection against credential theft

Team Development and Training

CTI Team Structure

Role	Responsibilities
CTI Analysts	Analyze threats and produce intelligence reports
SOC Analysts	Monitor alerts and investigate incidents

Incident Responders	Contain and recover from attacks
Compliance Officers	Ensure HIPAA compliance
Threat Hunters	Proactively identify hidden threats

Training Requirements

- Ransomware analysis training
- Threat hunting exercises
- Incident response simulations
- HIPAA security awareness
- Phishing detection training

Capability Evaluation

Current Capabilities

- Endpoint detection through CrowdStrike
- Threat feed integration
- Backup and recovery systems
- Security monitoring through SIEM

Capability Gaps

- Limited threat hunting maturity
- Incomplete automation of intelligence workflows
- Need for additional phishing simulations
- Limited third-party risk monitoring

Future Investments

- Security automation tools
- Advanced behavioral analytics
- Expanded threat intelligence subscriptions
- Additional CTI staffing

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6. Communication and Collaboration Channels

Internal Communication Channels

Channel	Purpose
Security Dashboards	Real-time threat visibility
Email Alerts	Immediate incident notifications
Weekly Security Meetings	Intelligence sharing and planning
Executive Briefings	Strategic risk communication
Collaboration Platforms	Cross-team coordination

External Collaboration

SecureHealth collaborates with:

- Healthcare Information Sharing and Analysis Center (H-ISAC)
- Government agencies such as CISA and the FBI
- Cybersecurity vendors and intelligence providers
- Industry cybersecurity communities

Benefits of Collaboration

- Early warning of healthcare-targeted attacks
- Shared threat intelligence
- Faster identification of ransomware campaigns
- Improved response coordination

Collaboration Challenges

Challenge	Improvement Strategy
Information-sharing delays	Automate intelligence sharing processes

Limited vendor transparency	Strengthen vendor security requirements
Communication silos	Increase cross-department collaboration meetings
Resource limitations	Invest in additional CTI staffing and automation

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7. CTI Effectiveness Measurement and Optimization

Key Performance Indicators (KPIs)

KPI	Purpose
Mean Time to Detect (MTTD)	Measure detection speed
Mean Time to Respond (MTTR)	Measure incident response efficiency
Ransomware Prevention Rate	Track blocked ransomware attempts
Phishing Click Rate	Evaluate employee awareness
Number of Detected IOCs	Measure intelligence effectiveness
Compliance Audit Results	Assess HIPAA compliance readiness

Continuous Improvement Process

SecureHealth continuously improves CTI operations through:

- Quarterly CTI strategy reviews
- Post-incident lessons learned meetings
- Threat feed updates
- Detection rule optimization
- Regular tabletop exercises and simulations

Optimization Activities

- Updating IOC databases
- Improving automated workflows
- Revising response playbooks
- Expanding intelligence coverage

Stakeholder Feedback

Stakeholder feedback is collected through:

- Security review meetings
- Incident debriefings
- Compliance audits
- Executive reporting sessions
- User awareness surveys

Application of Feedback

Feedback is used to:

- Improve intelligence reporting quality
- Enhance detection capabilities
- Refine incident response procedures
- Prioritize future CTI investments

This feedback-driven approach ensures SecureHealth’s CTI strategy remains adaptive, proactive, and aligned with evolving ransomware threats and healthcare security requirements.